



# Sustainable Development Verified Impact Standard

## IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA



Document Prepared by TEM Cookstoves Pty Ltd

|                      |                                                                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Title        | Improved Cookstove Grouped Project in Papua New Guinea                                                                                                                                                                                        |
| Version              | 1.0                                                                                                                                                                                                                                           |
| Date of Issue        | 14-January-2026                                                                                                                                                                                                                               |
| Project Location     | Papua New Guinea                                                                                                                                                                                                                              |
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|                                            |                                                                                                                                        |
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| <b>Project Lifetime</b>                    | 23 September 2024 to 22 September 2034; 10-year lifetime                                                                               |
| <b>Monitoring Period of this Report</b>    | 01 January 2025 – 31 December 2025                                                                                                     |
| <b>History of SD VSta Status</b>           | SD VSta Registered on 14/06/2024 <sup>1</sup>                                                                                          |
| <b>Other Certification Programs</b>        | VERRA Verified Carbon Standard (VCS) Project 4416                                                                                      |
| <b>Expected Future Assessment Schedule</b> | Verification anticipated in 2026                                                                                                       |

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<sup>1</sup> <https://registry.verra.org/app/projectDetail/SDVISTA/4416>

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# 1 SUMMARY OF SDG CONTRIBUTIONS

**Table 1: Summary of SDG Contributions**

| Row number | Quantitative Project Contributions during Monitoring Period                                                                                                                                                                                                                                                                                                                                                                   | Contributions during Project Lifetime                                                                                                                                                                                                                                                                                                                                                         | SDG Target | SDG Indicator                                               | Net Impact on SDG Indicator | Section Reference | Claim, Asset or Label |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------|-----------------------------|-------------------|-----------------------|
| 1)         | <p>For this current monitoring period, an additional <b>497</b> traditional cookstoves were replaced with fuel efficient Improved Cookstoves (ICSs) in the project area, with a focus in the Southern Highlands Province. The total number of operational cookstoves is <b>19,703</b>.</p> <p>The savings per household per day is estimated to be 1.04<sup>2</sup> Kina for households that purchase their own firewood.</p> | <p>The efficiency of changing from a traditional cookstove to a more efficient one will result in the consumption of lesser firewood that is usually purchased at high prices,<sup>3</sup> hence reducing the economic burden of families that buy firewood under the project activity.</p> <p>The project activity has replaced <b>19,703</b> traditional cookstoves with fuel efficient</p> | 1.1        | Goal 1: Amount of money saved by families that buy firewood | Increase                    | 3.2 #1            | Claim                 |

<sup>2</sup> The calculation method is demonstrated in the SD VISTa Monitoring Plan Spreadsheet made available to validators.

<sup>3</sup> ACIAR 2013\_Promoting diverse fuelwood production systems in Papua New Guinea

|    |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                    |     |                                                                                                                                                 |          |        |       |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|-------|
|    | <p>This is consistent with the responses from the monitoring survey, where households buying firewood reported an average savings of 5-10 Kina per week.</p> <p>By proportioning the number of households that purchase their firewood from the monitoring survey, <b>107,027 Kina<sup>7</sup></b> is saved during this period across all households.</p>                                                               | <p>ICSSs, leading to savings of 1.04 Kina per household per day.</p> <p>By proportioning the number of households that purchase their firewood from the monitoring survey, <b>164,917 Kina<sup>7</sup></b> is saved during the project lifetime.</p>                                                                                                               |     |                                                                                                                                                 |          |        |       |
| 2) | <p>Emissions of PM2.5 were estimated to be reduced to below the baseline emission level by 81.24% to 177 mg/MJ<sup>4</sup> in project households for newly distributed ICSSs, based on estimated from literature for the same stove model.</p> <p>Overall, this has resulted in a total decrease of <b>1,328.06 tonnes<sup>5</sup></b> of PM2.5 during this monitoring period based on the theoretical reduction of</p> | <p>Contribute to improved health and well-being brought about by reduced levels of fine particulate matter (PM<sub>2.5</sub>) emissions within households below the baseline emission level of 943 mg/MJ delivered<sup>6</sup>.</p> <p>Overall, a total decrease of <b>1,572.85 tonnes</b> of PM2.5<sup>7</sup> has been observed during the project lifetime.</p> | 3.9 | Goal 3: Reduction of fine particulate matter (PM2.5) emissions within households to reduce air pollution within households in the project area. | Increase | 3.2 #2 | Claim |

<sup>4</sup> [http://catalog.cleancookstoves.org/stove\\_details.html?stove\\_id=stove\\_AW0UJ](http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_AW0UJ)

<sup>5</sup> The calculation method is demonstrated in the SD VISTA Monitoring Plan Spreadsheet made available to validators.

<sup>6</sup> [http://catalog.cleancookstoves.org/stove\\_details.html?stove\\_id=stove\\_A8WGMJX](http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_A8WGMJX)

<sup>7</sup> The calculation method is demonstrated in the SD VISTA Monitoring Plan Spreadsheet made available to validators.

|    |                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |                                                                                                                        |          |        |       |
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|    | <p>baseline emissions due to the use of ICSs<sup>2</sup> as compared to traditional cookstoves<sup>3</sup>.</p> <p>Additionally, this is further supported by the Ground Monitoring Survey in which 100% of households reported observing only thin and light smoke and 94% of households reporting a decrease in intensity of the smoke compared to the baseline scenario.</p>                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |                                                                                                                        |          |        |       |
| 3) | <p>Non-formal education in the form of vocational and relevant skills was provided to <b>30</b> local individuals with respect to issues related to climate change, with specific skill building in operations and surveying activities related to ICS distribution and its monitoring under VCS have been provided during this monitoring period. All staff met the minimum passing rate of the quizzes given to them.</p> | <p>Contributed to increasing vocational and relevant skills of local individuals by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to ICS distribution and its monitoring under VCS.</p> <p>During the project lifetime, a total of <b>30</b> local staff were trained in training courses.</p> <p>These training courses were conducted annually for all the staff to make sure that the knowledge transferred is up</p> | 4.3 | Goal 4.1: Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex | Increase | 3.2 #5 | Claim |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |                                                                                                                      |          |        |       |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------|----------|--------|-------|
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>to date and the delivery of the project remains on a high standard.</p> <p>At the end of each training session, staff sat for an informal quiz to test their understanding of the training material. The measurement of improvement of learning outcomes and skill development will be based on the passing rate of the quiz.</p> <p>The minimum passing rate is at least 70%, and if a staff falls below the mark, they will have to re-sit the quiz within the same day.</p> |     |                                                                                                                      |          |        |       |
| 4) | <p>During this monitoring period, a total of <b>30</b> local staff were trained to conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and project activity. This is alongside training on how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle.</p> <p>The local staff will be annually trained to make sure that the</p> | <p>Contributed to increasing the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs, and entrepreneurship.</p> <p>A total of 30 local staff and personnel were trained during the project lifetime to conduct regular monitoring of the project sites for additional monitoring and</p>                                                                                                                  | 4.4 | Goal 4.2: Number of youth and adults with technical and vocational skills, by type of skill within the project area. | Increase | 3.2 #5 | Claim |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |                                                                                            |          |        |       |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------------------------------------------------------------------------------------|----------|--------|-------|
|    | knowledge transferred is up to date and the delivery of the project remains of a high standard.                                                                                                                                                                                                                                                                                                                        | maintenance of the ICSs and project activity. This is alongside training in how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle. The local staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains of a high standard.                                                                                                                                                         |     |                                                                                            |          |        |       |
| 5) | <p>During this monitoring period, the proportion of time spent cutting, collecting and carrying firewood from trees far away from households was monitored to be now <b>1.78% of a day</b> for people staying in the project area, compared to <b>6.25%</b> in the baseline scenario.</p> <p>This decrease from the baseline scenario results in total time saved on un-paid work to be <b>1.07 hours per day</b>.</p> | <p>The project will reduce women and children's drudgery through time savings in reducing time spent on cutting, collecting, and carrying firewood from trees far away from households.</p> <p>Monitored at <b>6.25% of a day</b> in the baseline scenario as per Tasman Environmental Market's Baseline Survey Report, participating households were now monitored to spend only <b>1.78% of a day</b> collecting firewood on average.</p> <p>As such, this decrease from the baseline scenario results</p> | 5.4 | 5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age, and location | Decrease | 3.2 #3 | Claim |



|    |                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |                                                                              |          |        |       |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------|----------|--------|-------|
|    |                                                                                                                                                                                                               | in a total time saved on unpaid work to be <b>1.07 hours per day</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |                                                                              |          |        |       |
| 6) | Out of the total <b>30</b> local individuals that has been trained and hired for this project, <b>8 (26.67%)</b> of them are female.                                                                          | <p>Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in the project.</p> <p>During the lifetime of the project activity, a total of <b>30</b> local individuals that has been trained and hired for this project, <b>8 (26.67%)</b> are female.</p> <p>Monitored as the fair and unbiased employment of women within the local team during the lifetime of the project. The project envisages that 50% of its employees will be female.</p> | 5.5 | Goal 5: Number of women in the project including managerial roles            | Increase | 3.2 #3 | Claim |
| 7) | As of this current monitoring period, an additional <b>497</b> traditional cookstoves were replaced with fuel efficient ICSs in the Southern Highlands project area, for a cumulative of <b>19,206 ICSs</b> . | The project contributes to increasing access to ICS technology with Burn cookstove installations in <b>19,703</b> households that have been using traditional three-stone fire, over the project lifetime.                                                                                                                                                                                                                                                                                                        | 7.1 | Goal 7: Number of people with primary reliance on clean fuels and technology | Increase | 3.2 #1 | Claim |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |                                                                                           |          |        |       |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------|----------|--------|-------|
| 8) | <p>During this current monitoring period, 6 additional people were onboarded for the project, for a total of <b>30 local staff</b> hired for the project (<b>2</b> full-time and <b>28</b> part-time). Out of the 30 staff, <b>26.67%</b> are females.</p> <p>A ratio of male to female employees is nearly <b>3:1</b></p> <p>All the staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains on high standards.</p> | <p>With the inception of the project activity, jobs will be created firstly for the distribution of ICSs. Personnel will also be trained in how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle.</p> <p>Further to that and aside from the monitoring done during the verification cycle, the project activity will also conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and project activity. In turn creating long term jobs for the local community, without prejudice of sex, age, occupation and persons with disabilities.</p> <p>During the lifetime of the project activity, a total of <b>2</b> full-time and <b>28</b> part-time local staff were hired. Out of the 30 staff, a total of <b>26.67%</b> are females.</p> <p>All the staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of</p> | 8.3 | Goal 8: Number of people formally employed, by sector and sex under the project activity. | Increase | 3.2 #4 | Claim |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------|----------|--------|-------|

|    |                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |                                                                                                    |          |        |       |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------|----------|--------|-------|
|    |                                                                                                                                                                                                                                                                                                                                                                         | the project remains of a high standard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |                                                                                                    |          |        |       |
| 9) | <p>In this monitoring period, out of the total of <b>30</b> staff, <b>28</b> personnel were employed part-time.</p> <p>These part-time employees were compensated with an average income of <b>(1,369.84 Kina / month)</b> and is pro-rated where part-time employment is relevant.</p> <p>This value considers age, sex, occupation and persons with disabilities.</p> | <p>Contribute to an increased economic income by providing fair compensation per month for the employees performing the jobs created as part of the project activity initially during deployment and later during project monitoring. Compensation will differ depending on the job scopes assigned to each formal employee employed under the project activity and will be sampled based on this distinction.</p> <p>Individuals employed under the project activity will likely be community members who are currently unemployed or working in the informal sector with an average income estimated as K439/month (Bonilla, et al., 2020).</p> <p>During the lifetime of the project activity, a total of <b>30</b> staff, including <b>28</b> part time</p> | 8.5 | 8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities | Increase | 3.2 #4 | Claim |

|     |                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                |      |                                                                                                                      |          |                       |                      |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------|----------|-----------------------|----------------------|
|     |                                                                                                                                                                                                                                     | <p>personnel were employed by the project.</p> <p>These part-time employees were compensated with an average income of <b>(1,369.84</b> Kina / month) and is pro-rated where part-time employment is relevant.</p>                                                                                                                             |      |                                                                                                                      |          |                       |                      |
| 10) | <p>Total emission reduction in the current monitoring period is <b>73,189 tCO<sub>2</sub></b>.</p>                                                                                                                                  | <p>Contribute to GHG emission reduction through a reduction of ~3.7 tCO<sub>2</sub>e per ICS due to the replacement of baseline stoves with Burn ICSs over the crediting period.</p> <p>Over the lifetime of the project activity, a total of <b>19,703</b> ICSs were distributed, generating a total of <b>86,635</b> tCO<sub>2</sub> eq.</p> | 13.0 | <p>Tonnes of greenhouse gas emissions avoided or removed</p>                                                         | Increase | VCS validation report | SD VISta-labeled VCU |
| 11) | <p>The project has resulted in saving <b>3.7 tonnes</b> of non-renewable biomass per year per ICS.</p> <p>Over this monitoring period, a total savings of <b>73,470</b> tonnes of biomass from <b>19,703</b> ICSs distribution.</p> | <p>Contribute an estimated reduction in removal of woody biomass to the tune of <b>3.7 tonnes</b> of firewood per ICS per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests.</p> <p>The above value is based on the difference between the</p>                            | 15.2 | <p>15.2.1 Progress towards sustainable forest management through the increase of above-ground biomass in forests</p> | Increase | 4.2 #1                | Claim                |

|  |  |                                                                                                                                                                                                                                                                                                                           |  |  |  |  |  |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|  |  | <p>amount fuel used with open-fire cooking (<math>BC_{b,i,y}</math>) and the amount of fuel used with the ICS (<math>BC_{p,j,k,y}</math>) based on the VM0050 Methodology.</p> <p>Over the lifetime of the project activity, a total savings of <b>86,916 tonnes of biomass</b> from <b>19,703</b> ICS installations.</p> |  |  |  |  |  |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|

## 2 PROJECT DESIGN

### 2.1 Project Objectives, Context and Long-term Viability

#### 2.1.1 Summary of Project Sustainable Development Objective(s)

Before the implementation of the project in Papua New Guinea, most of the local people in the island used non-renewable woody biomass for their fuel needs, for cooking with open fire or three-stone fire. The project activity will distribute fuel-efficient ICSs to replace the baseline cookstoves in households. Of which the ICS disseminated through this project will replace open-fired wood-fuelled cooking methods.

The project has been enabling and enhancing households to achieve several Sustainable Development Indicators (SDI):

The project provides ICSs with high thermal efficiency to reduce the consumption of firewood which helps achieve efficient use of an important type of nature resources, non-renewable biomass. This in turn impacts the estimated reduction in removal of woody biomass to the tune of ~3.7 tonnes of firewood per ICS per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests. This value is estimated from the difference between the amount fuel used with open-fire cooking ( $BC_{b,i,y}$ ) and the amount of fuel used with the ICS ( $BC_{p,j,k,y}$ ) based on the VM0050 Methodology. (SDI15.2.1)

The project activity aims to replace a total of 100,000 traditional cookstoves with fuel efficient ICSs. The efficiency of changing from a traditional cookstove to a more efficient one will result in the consumption of lesser firewood that is usually purchased at high prices. Hence reducing the economic burden of families under the project activity. (SDI Goal 1)

The distribution of ICSs will also introduce and increase access of the local population to ICS technology with Burn cookstove installations in around 100,000 households that have been using traditional open fire cooking, over the project lifetime. (SDI Goal 7)

It will also contribute to the improved health and well-being brought about by reduced levels of fine particulate matter ( $PM_{2.5}$ ) emissions within households below the baseline emission level of 943.30 mg/MJ. (SDI Goal 3)

With the inception of the project activity, jobs will be created firstly for the distribution of the ICSs. Further to that and aside from the monitoring done during the verification cycle, the project activity will also conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and project activity. This creates long term jobs for the local community, without prejudice of sex, age, occupation, and persons with disabilities. TEM envisions that 50% of the project's employees will be female. (SDI Goal 8 and SDI8.5.1)

With the creation of jobs, the project proponent will also carry out training for vocational and relevant skills of local individuals by providing non-formal education and training on issues

related to climate change, with specific skill building in operations and surveying activities related to ICSs distribution and its monitoring under VCS. (SDI Goal 4.1 and SDI Goal 4.2)

All in all, this contributes to an estimated average annual GHG emissions reduction of 3.7 tCO<sub>2</sub>e per ICS due to the replacement of baseline stoves with Burn Cookstoves over the crediting period. (SDG 13)

### 2.1.2 Description of the Project Activity

The Fuel-Efficient Cooking in Papua New Guinea project involves distribution of fuel-efficient ICSs in Papua New Guinea (PNG). The ICS disseminated through this project replaces open-fired wood-fuelled cooking methods.

The project is undertaken by TEM Cookstoves Pty Ltd (together with its subsidiaries, collectively mentioned as “**TEM**” in this document), during which 100,000 ICSs will be distributed free-of-charge to households in project area. TEM and its subsidiaries are also dedicated in creating additional value through this project, enhancing the local community’s health, well-being, and understanding and awareness of climate change and sustainable development.

The primary activity under this grouped project is the installation of ICSs.

**Improved Cookstoves (G3 Burn rocket stove (also known as Kuniokoa or Ecoa wood cookstove) with adjustable pot skirt):**

The primary project activity is the distribution and installation of the G3 Burn rocket stove, utilizing an air-cooled design in order to keep the inside of the combustion chamber hot and the outside cool, increasing durability, performance, and efficiency. Increased efficiency in burning woody fuel would mean less wood being used during the combustion process, encouraging local household to transition away from traditional open fire cooking to cleaner, more efficient cooking solutions.

#### **Training:**

In terms of training, TEM conducted a two-day course<sup>8</sup> for its local staff on climate change including the impacts of household pollution on health and well-being, as an introductory background to the efficient cookstove intervention. Training generates positive impacts on community groups by enabling local community members to build alternative skills and build capacity. These interventions are expected to advance socio-economic status, generate greater community capacity for sustainable livelihoods, and increase climate resilience.

During ICS distribution, these trainers assist stakeholders in understanding:

1. How the project activity works and its associated socio-economic benefits

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<sup>8</sup> Training course deck available to VVBs on request. This two-day training is only for new hires, existing hires will undergo a one-day refresher course before being sent for any field work.

2. How to use the ICS
3. End-User Agreement between TEM and End-Users

### 2.1.3 Implementation Schedule

| Date         | Milestone(s) in the Project's Development and Implementation                                                                                    |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-Jan-2023  | Consultation with Climate Change and Development Authority (CCDA) on Project Design and Local Stakeholder Consultation                          |
| 31-Jan-2023  | Consultation with PNG Forestry Authority (PNGFA) on Project Design and Local Stakeholder Consultation                                           |
| 13-Mar-2023  | Local Stakeholder Consultation in Southern Highlands Province                                                                                   |
| 15- Mar-2023 | Local Stakeholder Consultation in Western Highlands Province                                                                                    |
| 17- Mar-2023 | Local Stakeholder Consultation in Jiwaka Province                                                                                               |
| 20- Mar-2023 | Local Stakeholder Consultation in Eastern Highlands Province                                                                                    |
| 30-Jun-2023  | Listing of Project under the SD-VISta                                                                                                           |
| 13- Jul-2023 | Listing of Project under the VCS Program                                                                                                        |
| 13-Aug-2023  | Undergoing validation (Public comment period closed)                                                                                            |
| 01-Jan-2024  | Estimated Project Start date: Project Activities commence with G3 Burn rocket stove (known as Kuniokoa Cookstove) distribution and installation |
| 14- Jun-2024 | Project Registration Approval under the VCS Program                                                                                             |
| 29-Aug-2024  | Project Registration Approval under the SD-VISta Program                                                                                        |
| 23-Sep-2024  | Project Start Date                                                                                                                              |
| 30-Sep-2024  | Start Date for Monitoring Period Commissioning of Batch 1                                                                                       |
| 31-Oct-2024  | Start Date for Monitoring Period Commissioning of Batch 2                                                                                       |
| 28-Mar-2025  | Start Date for Monitoring Period Commissioning of Batch 3                                                                                       |
| 05-Aug-2025  | Verification and issuance of credits from first monitoring period (23-Sep-2024 to 31-Dec-2024)                                                  |

### 2.1.4 Project Proponent

|                            |                                                                             |
|----------------------------|-----------------------------------------------------------------------------|
| <b>Organization Name</b>   | TEM Cookstoves Pty Ltd                                                      |
| <b>Role in the Project</b> | Project Developer                                                           |
| <b>Contact Person</b>      | Dr. Gabriel Machovsky Capuska                                               |
| <b>Title</b>               | Director of International Projects                                          |
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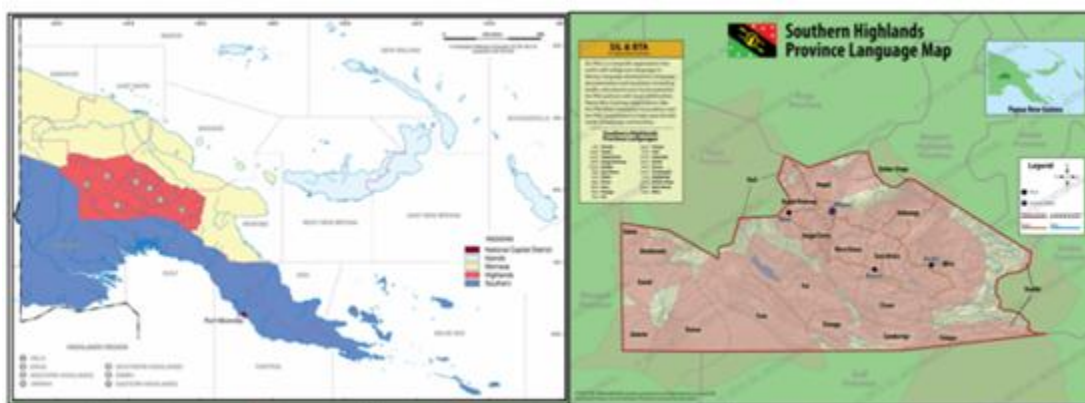
### 2.1.5 Other Entities Involved in the Project

TEM is the sole entity involved in the project.

|                     |                |
|---------------------|----------------|
| Organization Name   | Not Applicable |
| Role in the Project |                |
| Contact Person      |                |
| Title               |                |
| Address             |                |
| Telephone           |                |
| Email               |                |

### 2.1.6 Project Location

The GPA will distribute ICSs to communities within the project boundary of PNG (Latitude: 6.3150° S Longitude: 143.9555° E). The first project instance monitored in this monitoring report is within the Southern Highlands Province (see *Figure 1 & Table 2*).



**Figure 1:** Map of Papua New Guinea including the Highlands Region (left) and map of Southern Highlands Province (right). Maps extracted from [www.pnglanguages.sil.org](http://www.pnglanguages.sil.org)

**Table 2:** The Geographic coordinates of the two provinces involved as the first project instances of the group activity in Papua New Guinea

| S/N | Location                    | Latitude  | Longitude   |
|-----|-----------------------------|-----------|-------------|
| 1.  | Southern Highlands Province | 6.4179° S | 143.5636° E |

### 2.1.7 Project Description Deviations

According to the Project description, monitoring of PM<sub>2.5</sub> was to be conducted through measurements taken in the household during sampling within the households' kitchen area during a cooking activity. Reliable PM<sub>2.5</sub> measurements in household environments are a technical challenge due to large spatial variabilities across households, differences in cooking practices, inconsistent data from low-cost sensors, difficulty in isolating indoor sources from outdoor infiltration, the need for long-term monitoring to capture seasonal variations, and the complexity of accurately capturing personal exposure due to movement within the house<sup>9</sup>. All of these can significantly impact the accuracy of PM<sub>2.5</sub> measurements within a household environment.

In the project description deviation, it was decided to use the results from the extensive standardised testing (using the ISO Protocol 19867-1:2018) available in the Clean Cooking Alliance database<sup>10</sup> for the specific model of the project stove, comparing the with the same parameters for open fire cooking. The comparison of supporting evidence from the database catalogue on clean cookstoves demonstrated a reduction in PM<sub>2.5</sub> in households to be by 81.24% from 943.40 mg/MJ<sup>11</sup> to 177 mg/MJ<sup>12</sup> for newly distributed ICSs. Hence, calculation on the decrease in PM<sub>2.5</sub> is possible based on this literature, project specific parameters and the number of ICSs distributed. A detailed calculation is available upon request.

In order to qualitatively account for and supplement evidence for the decrease in PM<sub>2.5</sub>, during the Ground Monitoring Surveys, questions on the type of smoke emitted during cooking: (1) Thin and Light, (2) Medium and Grey and (3) Thick and Dark was surveyed to households. This was compared to the baseline where only Thick and Dark smoke was observed, to determine if there was a decrease in intensity of smoke produced.

Results from this qualitative survey conducted for the monitoring period showed that 100% of households reported observing only thin and light smoke and 94% of households reported a decrease in intensity of the smoke compared to the baseline scenario.

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<sup>9</sup> Assessing household fine particulate matter (PM<sub>2.5</sub>) through measurement and modeling in the Bangladesh cook stove pregnancy cohort study (CSPCS), M.M. Rahman et. al, 2023, <https://www.sciencedirect.com/science/article/abs/pii/S0269749123015701>

<sup>10</sup> Clean Cooking Alliance Database: <http://catalog.cleancookstoves.org/>

<sup>11</sup> [http://catalog.cleancookstoves.org/stove\\_details.html?stove\\_id=stove\\_A8WGMJX](http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_A8WGMJX)

<sup>12</sup> [http://catalog.cleancookstoves.org/stove\\_details.html?stove\\_id=stove\\_AW0UJ](http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_AW0UJ)

### 2.1.8 Threats to the Project

**Table 3:** Nature-induced threats

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Threat:</b>   | Continued deforestation and degradation make firewood an untenable source of biomass.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Solution:</b> | <p>As much as the efficiency of the cookstoves coupled with the imminent scarcity of the wood fuel would support the use of G3 Burn rocket stove (also known as Kuniokoa Cookstoves) as proposed by TEM for this project, there is the potential that as the population continues to expand and there may be a point when firewood becomes an untenable source.</p> <p>In response to this, TEM's in country team will train end users to maximise the use of small twigs and branches, crop residues near the local communities they live.</p> |

**Table 4:** Human-induced threats

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Threat:</b>   | End users' lack of maintenance of ICS resulting in either malfunctioning of the ICS as designed or lack of use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Solution:</b> | <p>TEM will conduct routine visits to end-user households at least twice a year in each main geographic areas that currently use the new G3 Burn rocket stove (also known as Kuniokoa), to identify potential defects in the stove's construction, maintenance, and sub-optimal use. Additionally, the end-users also have access to a grievance redress mechanism (Section 2.2.7) to report damaged cookstoves to the project staff.</p> <p>To reduce this proposed treat, TEM's in-country team will hire, train, and supervise users that will involve visiting the end-user's houses periodically to monitor the use of the new ICS.</p>                                  |
| <b>Threat:</b>   | Failure of Behaviour Change resulting in high levels of non-adoption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Solution:</b> | <p>TEM will continuously research the impacts of behaviour change for successful adoption. Study outcomes, whether formal or informal, will be included in future training and community engagements to continue shaping knowledge, and enhancing positive peer-to-peer influence.</p> <p>Planned community incentivisation mechanisms to reduce the use of inefficient baseline devices and practices are explained below:</p> <ul style="list-style-type: none"> <li>- Field assistants known as 'Cookstove Ambassadors' will actively interact with end-users from their allocated jurisdictions to encourage the use of ICSs and continuously foster awareness</li> </ul> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>on the benefits of using ICS over open fire cooking methods. The increase in adoption rates will be monitored through surveys inquiring about the rate of stove stacking within the jurisdiction.</p> <ul style="list-style-type: none"> <li>- Cookstove Ambassadors will also present successful stories through knowledge sharing sessions with the event's participants.</li> </ul> <p>Additionally, TEM will work with Cookstove Ambassadors seeking successful testimonials from end-users benefiting from clean cooking practices through the use of ICSs that can be converted into short form videos to capture broader social media audiences.</p> <p>Non-adopters, where identified, will also be engaged to understand the potential barriers preventing them to adopt the use of the ICS. These outcomes and conclusions will be useful to improve future training events with the goal of avoiding miss-information.</p> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                  |                                                                                                                                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Threat:</b>   | ICS Projects and distributions are not shown to have an increased impact on household uptake.                                                                                                                                                                                                                         |
| <b>Solution:</b> | Time savings and increased health benefits are co-benefits from the adoption of ICSs will be incentives to increase their uptake. In response to this, TEM will conduct trainings highlighting how time saving, health benefits, previous users' positive experiences are important factors for large scale adoption. |

### 2.1.9 Benefit Permanence

To ensure the project's long-term viability, the permanence of the individual ICS and other program benefits, TEM will take the following long-term approach:

- 1) Periodical monitoring of the end users to ensure the regular use of ICSs. To guarantee the success of this process our local staff will conduct visits to the communities multiple times per year. These teams will be provided with transportation and tablets that meet the physical mobility and remote connection requirements of their roles.
- 2) Conduct training sessions for the local communities to increase the awareness levels of these communities regarding the long-term environmental, health, and resiliency benefits aligned with the adoption of ICSs. The understanding of these holistic benefits to the stakeholders is valuable for ensuring long-term sustenance and maintenance of project activities and achieving the intended outcomes.

- 3) TEM and the Western Pacific University (WPU) signed a Memorandum of Understanding (MoU) (see file titled: VCS4416\_MoU\_TEM and WPU\_PNG\_signed 21020224.pdf) to collaborate on the development of supporting environmental and sustainability activities directed to educate and train local students in topics related to sustainability, climate change impacts and mitigation strategies. TEM interacts with the WPU through various engagement sessions on the opportunity to educate local youth on the benefits of clean energy and clean cooking practices and to better understand the extent of the PNG ICS project as a mitigation strategy for climate change. As stated in the MoU, the main goal is to involve the local communities in the proposed project activities, creating a sense of ownership amongst the people, and inspiring behavioural change from a young age through formal education.

The project also rolled out the “SME<sup>13</sup> Mamas” program, branded as “Cooking for Change – *Klinpela Kuk Gutpela Laif*”, a women-led initiative that empowers unemployed mothers (mamas) to set up a microbusiness selling pancakes using ICSs. It aims to provide financial literacy and income opportunities for women, nutritious meals to school children, and promote clean cooking practices. The program includes training on cookstove safety, hygiene, recipe mastery, financial literacy, and marketing. TEM supports them with a small seed funding to purchase essential ingredients (“basket”) to develop a microbusiness selling pancakes. This initial financial assistance removes a critical barrier to entry, allowing women to start operations without incurring debt or relying on informal credit. By ensuring access to basic resources, TEM helps to create a foundation for sustainable income generation and empowers women to take ownership of their enterprises from the onset. Once a mama breaks even through the sales from her stall, they pay forward the “basket” of products received to help another mama from the community to start their own initiative. This rotating “basket cycle” model ensures scalability and sustainability, while mentorship and recognition programs build confidence and community bonds.



Figure 2: A 'mama' receiving a cooking 'basket' from TEM staff (left)  
A 'mama' in her stall selling pancake to a customer (right)

<sup>13</sup> SME refers to small-medium enterprises

Beyond seed funding, TEM delivers comprehensive financial literacy and business training tailored to the needs of these women. The training covers budgeting, record-keeping, pricing strategies, and marketing techniques, equipping participants with the skills required to manage and grow their businesses effectively. This holistic approach not only strengthens individual capacity but also fosters long-term resilience and scalability within the program, ensuring that women can transition from subsistence activities to thriving microenterprises.

These efforts will ensure that over the duration of the project the end users will have a positive behavioural change through the improved awareness, education and first-hand experience of the health and economic benefits over traditional open fired cooking, which will in turn lead to the continued use of ICSs even after the period of project activity. Evidence for benefit permanence stated above is further discussed in section **Error! Reference source not found.**

## 2.2 Stakeholder Engagement

### 2.2.1 Stakeholder Consultation and Adaptive Management

During this monitoring period, stakeholder consultations presented are focused on the provinces where distribution of ICSs took place.

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date of stakeholder consultation</b> | Southern Highlands Province: 13 March 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Stakeholder engagement process</b>   | <p>Prior to the LSCs, TEM utilized several media outlets to advertise the event to the provinces. These efforts were fulfilled using:</p> <ul style="list-style-type: none"> <li>• Newspaper Advertisements</li> <li>• Letters of Notification to Provincial Governments</li> <li>• Radio Broadcasts</li> <li>• Invitation Letters to Churches</li> <li>• Social Media Posting</li> <li>• Webinar</li> <li>• Email to PNG Australia Alumni Association (PNGAAA) members</li> </ul> <p>All advertising outlets were run in a staggered format with the aim of keeping stakeholders informed and aware of the LSCs that would be happening within their provinces. For the exact dates of advertisements being published, please refer to the PD Appendix II: LSCs Outreach.</p> <p>The stakeholders engaged in this project were people or groups of people that were directly or indirectly affected by the project as well as those who were interested in the project and or had the ability to influence its outcome either positively or negatively. They include:</p> <ul style="list-style-type: none"> <li>• Representatives from the provincial administration</li> </ul> |



- Representatives from Women Groups
- Representatives from Churches
- Ward councilors
- Heads of households both male and female
- Key informants such as teachers, community health workers etc.
- Improved Cookstove (ICS) users (SHP only)
- Youth leaders and
- Educated elites (lecturers, nursing officers, Health Extension Officers, Police personals etc.)

#### Southern Highlands Province LSC

A total of 159 people attended this LSC on the 13th of March 2023 held at Kumin Catholic Mission in Mendi, SHP. Including those who commuted into Mendi from various districts, there were public servants such as nursing officers, police, teachers and church elders and pastors present during the consultation.



Figure 3: A demonstration by Ms. Magret Yangpin and Mr. Richard Manda – SHP Field Assistants.

The LSC was facilitated by Mr. Michael Kisombo, the project Manager with Simi Oa Consultancy Services who was engaged by TEM, did a power point presentation after welcoming and acknowledging all the stakeholders that were present. The PowerPoint slides were prepared in English, and explained in tok pidgin, the common language where most could understand well as those participated came from different local dialects.

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>Attendees were briefed on the purpose of the meeting, and informed of the:</p> <ul style="list-style-type: none"> <li>- Carbon cycle and the important role it played within their forests.</li> <li>- Findings of the Baseline Study conducted in the Highlands region including SHP and WHP (file titled: VCS4416_TEM 2025_Cookstoves Highlands region report_v3.pdf)</li> <li>- The Verra registration process</li> <li>- The role of carbon finance in the project activity</li> <li>- Indicative timeline and implementation plan of the project</li> </ul> <p>Two field assistants as seen in Figure 3 were then invited to demonstrate and to explain how the ICS is used using the sample that was exhibited during the LSC. That was further reinforced by one of the users who was also invited to give her experience of how she is currently using the ICS and explained the benefits her family is experiencing.</p> <p>After the presentation, stakeholders were given the opportunity to ask questions and provide their reflections. For the summary of questions asked and subsequent answers provided by TEM, please refer to Appendix III: Stakeholder Feedback.</p> <p>After the discussions, there were about 29 feedback forms that were received through the feedback forms. These feedback forms may be made available to validators upon request.</p> |
| <b>Consultation outcome</b>  | <p>There was no negative feedback received during the LSC. Most questions aimed to seek clarity on how a more efficient ICS could reduce carbon in the atmosphere, when the project could begin, and if more ICSs could be distributed. For a complete list of questions and feedback, please refer to the PD Appendix III: Stakeholder Feedback.</p> <p>For more information, a Local Stakeholder Consultation Report is available upon request of the validator.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Ongoing communication</b> | <p>During the Local Stakeholder Consultations, it was discussed how stakeholders may keep in contact with TEM's in country project manager, Michael Kisombo. During which, the following solutions were offered:</p> <ol style="list-style-type: none"> <li>1. Reaching out to the in-country Team during the twice a year visits to maintain the ICSs.</li> <li>2. Reaching out to TEM through the project's dedicated email.</li> <li>3. Reaching out to TEM through the project's website.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p>Stakeholder consultations conducted during the monitoring period, please refer to Appendix I: Stakeholder Consultations</p> <p>The following website of the project activity was available to the stakeholders:<br/> <a href="https://www.tasmanenvironmental.com.au/pngcookstoves/">https://www.tasmanenvironmental.com.au/pngcookstoves/</a></p> <p>The following email addresses will be also used for inquiry or comment on the project: <a href="mailto:pngcookstoves@tem.com.au">pngcookstoves@tem.com.au</a></p> <p>TEM PNG Country Manager: Michael Kisombo, +675 7262 3864, <a href="mailto:michael.kisombo@tem.com.au">michael.kisombo@tem.com.au</a>.</p> |
| Stakeholder input | Please refer to the PD Appendix III: Stakeholder Feedback.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

For more information on stakeholder consultations conducted during the monitoring period, please refer to Appendix I: Stakeholder Consultations.

Overall, TEM engaged with ICS end-users during visits to register the households and to give out the ICSs. In total, TEM has registered and distributed an additional 497 ICSs during this monitoring period, bringing the cumulatively deployed ICSs to 19,703. In addition, monitoring surveys and training of new staff was also conducted. In addition, monitoring surveys and training of new staff were also conducted where the results are further described section **Error! Reference source not found.**

### 2.2.2 Anti-Discrimination

TEM will distribute ICSs to local people who uses traditional low efficiency cooking solutions without distinction in genders, races, religions, educational backgrounds, or any other aspects. TEM is dedicated to the elimination of discrimination, and as jobs are created during the implementation of this project, the company will ensure that women, persons with disabilities, and minority will have equal chance to get these jobs.

All TEM Employees adhere to the TEM Employee Handbook which instils the importance of preventing Harassment (Section 13) and Equal Opportunities and Anti-Discrimination (Section 14). This handbook informs and ensures that all employed under TEM ensures that any form of harassment or bias in project operations is prevented by clearly laying out the definitions of harassment and discrimination, and its consequences.

These policies were enforced throughout the monitoring period.

### 2.2.3 Worker Training

The implementation of the project needs plenty of local people to participate in production, distribution or use steps. They will get relevant skills and sustainable development and global

citizenship education through training by project proponent. Besides technical skills, the workers will be trained about the management structure, regulations, and worker rights.

TEM organised a training course<sup>14</sup> between 15-Jan-2025 to 17-Jan-2025 for the in-country team to learn about monitoring techniques including survey methods, data record and analysis. The monitoring plan was carried out by qualified personnel trained for quality assurance and quality control.

To ensure long-term skill retention, refresher training courses for the field team are conducted prior to each monitoring on 05-Jan-2026 for the previous monitoring surveys and on 05-Jan-2026 for the current monitoring survey. To ensure continuity in this, a refresher will be always conducted prior to any monitoring surveys in the future.

The field team was also trained on how to conduct Kitchen Performance Tests (KPTs) and subsequently carried out the KPTs successfully for all batches between July and August 2025. The field team was also provided with training on ISO Water Boiling Test protocol from the cookstove manufacturer, Burn Manufacturing, on 05-Nov-2025 and carried out WBTs along with the technical staff from Burn.

For inclusivity, training such as the technical skills and education on regulations were provided for marginalized and vulnerable group as 26.67% of the individuals trained were female.

#### 2.2.4 Equal Work Opportunities

TEM is dedicated to make sure that all stakeholders, no matter their genders, races, religions, educational backgrounds, or any other aspects, will be given an equal opportunity to fill all work positions. Recruitment advertising for the local villages will be given before the hiring of the full-time and part-time staff for the project. The recruitment advertisement will be provided in Pidgin English for them to understand.

These policies were enforced throughout the monitoring period and hiring was done fairly in partnership with the recruitment agency, Vanguard, which was in charge of recruiting, interviewing and contracting full-time employees according to TEM's Job Description and Equal Opportunities & Anti-discrimination Policy found in the TEM Employee Handbook, Section 14.

During the monitoring period, it was noted that 26.67% of the individuals hired were female.

#### 2.2.5 Workers' Rights

TEM provides a labour contract with each in-country personnel consistent with PNG labour legislation. Before the workers sign it, the terms of the contract are explained explicitly to make sure that they could understand their rights and obligations. The contract terms conform with and uphold the principles and rights of work addressed in the Core Labour Conventions of the

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<sup>14</sup> Training course deck available to VVBs on request. This two-day training is only for new hires, existing hires will undergo a one-day refresher course before being sent for any field work.

International Labour Organization (ILO), as well as comply with national and local labour laws within Papua New Guinea.

During the monitoring period, each worker is placed in teams where there is a hierarchy for reporting. During the hiring process, it is known to the workers (regardless of position) that reporting of any labour rights violations or grievances is possible through reporting it to a superior.

These policies were enforced throughout the monitoring period.

### 2.2.6 Occupational Safety Assessment

To mitigate potential occupational safety hazards, TEM makes the effort to contract workers who have lived in the community for several years, thus assisting in mitigating the occupational hazards through community familiarity, language fluency, and being native to the culture. This understanding of traditional values, respect, and working environment in the communities TEM serves to support reduction of safety hazards.

Risks are assessed and mitigated where possible and workers, stakeholders, and others directly involved with the projects are made aware of the potential risks and opportunities to mitigate where possible, yet it should be noted that employment or use of the ICS does not increase the safety risks that are already present in the landscape of the country.

TEM ensures safe and appropriate working conditions for all its employees through the development and consultation of its Project Risk Assessment. Within this assessment sheet, risks related to transportation and security are identified for stakeholders including TEM staff, the consequences of risks to occur and their likelihood of it happening is stated via an impact score. Then, mitigation actions for these risks are included to bring the residual risk down to acceptable levels.

Additionally, all personnel who travel into the project boundary must adhere to the tailored security travel plans that are developed with clear emergency protocols to guarantee their safety. The travel plan includes details for Remote Check-Ins with a designated Check-In person, full travel itinerary, emergency contact details, etc., and must be pre-approved by TEM's Director of International Projects before they are circulated to all parties participating in the trip.

### 2.2.7 Feedback and Grievance Redress Procedure

No grievances were reported during the monitoring period.

The details on procedure for providing feedback and raising grievances have been made publicly available to all stakeholders on the project webpage on TEM's website that can be accessed at the following link: <https://www.tasmanenvironmental.com.au/pngcookstoves/>

Additionally, a Grievance Redress Mechanism (GRM) Standard Operating Procedure (SOP) is prepared by TEM which aims to present a set of guidelines for designing and implementing different mechanisms that can address a variety of grievances arising when installing the ICSs in PNG during the crediting period of the project activity. The guidelines are prompt and effective and are timely to inform the complainants of the actions taken to resolve issues raised. The

channels for reporting the grievances and the process of categorising common grievances are provided below.

**Table 5:** Channels for Grievance Reporting

| Grievances Reporting Channel             | Grievance Reporting Mechanism                                                                      |
|------------------------------------------|----------------------------------------------------------------------------------------------------|
| Physical walking in                      | Complaints to TEM PNG Office and/or its staff                                                      |
| Social media and online                  | Complaints made online to TEM PNG Office, and/or in country Project manager                        |
| Phone call                               | Phone calls to Field Assistance, and/or Project manager                                            |
| Verbal through community representatives | Complaints through community representatives who then channel through the organizational structure |

**Table 6:** Process of organizing common grievance

| Type of Grievance                               | Scope                                                   | How will be facilitated                                        | Who will handle                  | Timeline     |
|-------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------|----------------------------------|--------------|
| Unfair distribution of ICS                      | Cookstoves unfairly distributed by project staff        | Gather information, investigate and refer to project lead      | In country Project manager       | 7 -14 days   |
| Burglary, sold or destroyed by natural disaster | Cookstoves were stolen, destroyed, or sold              | Gather information, investigate, and refer to project lead     | Project manager and project lead | 14 – 21 days |
| Household missing out on the distribution       | Cookstoves missed targeted households                   | Investigate and refer to project lead for possible re-issue    | As above                         | 5 – 7 days   |
| Damaged cookstoves                              | Cookstoves issued were damaged or have technical issues | Investigate and refer to project lead for possible replacement | As above                         | 5 -7 days    |
| Complaints raised by project staff              | Unfair decisions related to pay or disciplinary actions | Gather information and refer to project lead                   | Project manager                  | 5 – 7 days   |

With reference to the GRM, all complaints and grievances will be registered and monitored by the in-country Project Manager supported by the field assistants. These complaints and grievances will be resolved using traditional and customary ways.

This document will be socialized and publicized to communities throughout the project location in their local language, Pidgin English.

During the monitoring period, no formal complaints were received or logged from the end-users in the form of physical walk ins, social media and online postings or phone calls, and verbally through community representatives.

Verbal feedback has been very positive, while one feedback was made in the presence of the validator during the site visit regarding soot from open fire cooking being able to maintain the waterproofing of the end-user's dwelling. However, as the outcome of using the project ICS in lieu of open fire cooking cannot be scientifically proven, this feedback has been recorded as informal feedback.

### 2.2.8 Stakeholder Access to Project Documentation

TEM informs the beneficiaries and their families that they are voluntarily participating in a project that distributes the G3 Burn rocket stove (also known as Kuniokoa cookstove) upon the household's receipt of the ICS. TEM also informs them of the need to track their successes and monitor key data, such as money spent on wood fuel and/or time spent collecting wood fuel, among other indicators. ICS end-users will also be made aware that to ensure the integrity of the project and programs, auditors, and cookstove ambassadors may visit the families up to 2 times after implementation to support with awareness and encouraging behavioural change.

As for project documents, the full project documentation will be uploaded on VERRA website, and the same ones will be publicized on the TEM website during the crediting period of the project at the following link:

<https://www.tasmanenvironmental.com.au/pngcookstoves/>

### 2.2.9 Information to Stakeholders on Assessment Process

To ensure all the stakeholders are aware of any assessments, including the site visits of assessors, TEM and its in-country team will inform them in advance (at least 7 days) through a combination of website link, phone, email and the following communication channels that are effective in the community.

#### 1. Community announcements through local newspapers and radio *toksaves*

Local newspapers: 'The National and Post courier', and radio shows like *Radio nek blong eagle* (Radio Western Highlands Province) have been used to inform about stakeholder meetings in the past and will be continued. These are paid advertisements those run as community announcements prior to meetings.

#### 2. Facebook Forums and WhatsApp groups

There are several Facebook sites and WhatsApp groups created to discuss developmental issues in each province. For instance, SHP Development forum, EHP Development forum, WHP development forum, among others. This has been utilised for previous stakeholder engagement program details including venues for community engagement, time and date and so forth using both English and tok pidgin (national language).

#### 3. Community meeting platforms and word of mouth

Church gatherings are significant places to inform stakeholders about the project. Community announcements will be prepared in Tok Pidgin and issued to Parish Priests, Pastors and Church elders who make the announcements during their church gatherings. As a significant population are illiterate, phone communications to village elders, ward councillors and project field assistants to spread the information are also an effective method to disseminate information.

The meetings of stakeholder with assessors will be facilitated at a venue informed in advance through the above channels and in the presence of an adequate interpreter (if required due to language restrictions) who would be a community member but not associated to the project.

## 2.3 Project Management

### 2.3.1 Avoidance of Corruption

TEM staff worldwide follows strict and robust rules and regulations regarding the avoidance of corruption, code of conduct, and business ethics.

All TEM employees adhere to the Anti-Bribery and Corruption Policy Laws to protect both themselves and the reputation of the company. Under this policy, TEM employees are not permitted to give, offer, promise, accept, request, or authorise a bribe, whether directly or indirectly or abuse entrusted power for private gain.

Any person or organization which violates these anti-corruption rules and regulations will have their engagement immediately terminated by TEM and its local partners.

### 2.3.2 Recognition of Property Rights

All ICSs included in this project activity were distributed to individual households free of charge. Before receiving their ICS, end-users will sign a carbon transfer agreement (i.e., End-User Agreement) with TEM, clearly stating that the ICS will belong to them, whereas the carbon credits generated from their use will belong to TEM.

### 2.3.3 Free, Prior and Informed Consent

The project is voluntarily implemented by TEM, and end-users are free to choose whether they will be involved in the project or not. Free, prior, and informed consent takes place before distribution through signing of the End User Agreement. In the agreement, end-users acknowledge that they have received the ICS, and they will transfer the rights of the carbon credits generated from the project to TEM.

### 2.3.4 Restitution and/or Compensation for Affected Resources

The project activity involves distribution of ICSs to individual households only and it will not affect any resources.

### 2.3.5 Property Rights Removal/Relocation of Property Rights Holders

The project activity involves distribution of ICSs to individual households only and it will not lead to any removal of property rights or relocation of property rights holders.

### 2.3.6 Identification of Illegal Activities

Theft and corruption may be identified during the implementation of the project. TEM and its local partners will avoid any corruption as per Section **Error! Reference source not found.** of this document. End-users will be reminded to keep their ICSs in safe places to reduce theft when they receive the ICSs.

### 2.3.7 Ongoing Conflicts or Disputes

Where applicable, describe activities and/or processes implemented to resolve conflicts or disputes.

The project activity involves distribution of ICSs to individual households only. There is no ongoing or unresolved conflicts or disputes over rights to lands, territories and resources and any disputes that were resolved during the last twenty years.

### 2.3.8 National and Local Laws and Regulations

The project aligns with several International and Papua New Guinean's laws, regulations and conventions. TEM, as the project developer, and its in-country team will follow and obey all these relevant laws, regulations and conventions.

These are stated below:

#### **International Laws, Regulations and Conventions**

- **United Nations Framework Convention on Climate Change (UNFCCC,1992)**

The aim of the UNFCCC is to achieve the "stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system".

The project has been implemented to reduce Green House Gas (GHG) emissions in an effort to combat climate change, and therefore the Project is compliant with the UNFCCC.

- **United Nations Declaration on the Rights of Indigenous Peoples Rights (2007)**

The Declaration delineates the individual and collective rights of indigenous peoples, their land, territories, resources, culture, identity, language, employment, health, education, and other issues. It also "emphasises the rights of indigenous peoples to maintain and strengthen their own institutions, cultures and traditions, and to pursue their development in keeping with their own needs and aspirations". It "promotes their full and effective participation in all matters that concern them and their right to remain distinct and to pursue their own visions of economic and social development".

The project is aligned with this United Nations Declaration by providing the local (indigenous) communities with the opportunity to engage, at their own free will, in economic development via



carbon finance, protection of their lands, territories and forest resources while also enhancing opportunities for other livelihoods, education, employment and healthcare. Furthermore, the project aims to improve the general way of life for current and future generations, including education on traditional ways of *life* and enhancing community by way of communication, meeting places and knowledge sharing.

- **Stockholm Declaration of the United Nations Conference in Human Environment (1972)**

The Stockholm Declaration states that it is each person's right to freedom, equality, adequate conditions of life; responsibility to protect the environment for present and future generations; to safeguard natural ecosystems and ensure the capacity of the earth to produce renewable resources is maintained, restored, and improved.

The project provides a mechanism whereby the local community and the Papua New Guinean Government can protect the ecosystems and environment of the Project Area, and lead to an improvement in the socio-economic conditions of the community. In addition, the renewable resources of the area will be maintained and possibly improved by the implementation of the project. The socio-economic well-being of local communities will also be improved through income and community development programs.

- **Rio Declaration on Environment and Development (1992)**

The 'Rio' Declaration aims for people to have a healthy and productive life in harmony with nature. It aims to ensure that development activities within national jurisdiction do not cause damage in areas beyond the national boundary. It also aims to eradicate poverty and encourage sustainable development and environmental protection.

The project aims to avoid GHG emissions within Papua New Guinea, which would have had an adverse impact on the global climate extending beyond the country's boundaries. The project also encourages an increase in sustainable livelihoods and education to combat poverty in the project area through sustainable development and environmental protection practices.

- **Paris Agreement (2015)**

The Paris Agreement's goal is to keep the rise in mean global temperature to below 2 °C above pre-industrial levels and preferably limit the increase to 1.5 °C to substantially reduce the effects of climate change, and for emissions to be reduced as soon as possible and reach net zero by the middle of the 21st century.

In alignment with this agreement, the project activities reduce GHG by avoiding carbon emissions.

### **Papua New Guinea's Laws, Regulations and Conventions**

- **Papua New Guinea's Constitution (1975)**

The Project is compliant with Papua New Guinea's Constitution, specifically the fourth goal of the Constitution which is "to ensure that the forest resources of the country are used and replenished for the collective benefit of all Papua New Guineans, now and for future generations".



The project aims to reduce the dependence of the people on firewood and in turn avoid deforestation.

- **Climate Change Management Act 2015, Amended (2022), and proposed Carbon Market draft regulation (2023)**

The Act established the Climate Change Development Authority (CCDA) and provided a framework for the development and implementation of measures in Papua New Guinea to combat climate change impacts in accordance with UNFCCC and other international agreements and programs. The project is aligned with the CCDA policies and the proposed carbon market draft regulations.

- **Forestry Act (1991)**

The Act is aimed at managing, developing and protecting Papua New Guinea's forest resources; maximising participation in and development of forest resources as renewable assets; utilising the forest resource to achieve economic growth, employment and industry; and encouraging scientific study and research to contribute to sound ecological balance.

The project assists Papua New Guinea in achieving its goals under the Act to protect and use forest resources through the reduction of firewood consumption and the development of carbon credits; while achieving economic growth through direct employment opportunities, education, and skills development in the project area.

- **Environmental Act (2000)**

The Environmental Act outlines the protection of the environment in accordance with the Fourth National Goal of the Constitution; promotes sustainable development of the environment and economy; states that the social and physical well-being of the people should be supported by the safeguarding the life-supporting capacity of the air, water, soil, and ecosystems for present and future generations, and avoiding, mitigating any adverse effects of activities on the environment.

*The project complies with requirements of the Act to promote sustainable development of the environment (protection) and economy (carbon credits, improved cooking) while supporting the social (training, community health) and physical wellbeing (healthcare) of the local communities. The project also reduces the deforestation through reduced demand of firewood.*

- **Investment Promotion Act (1992)**

The Papua New Guinea Investment Promotion Authority (IPA) was established by the Investment Promotion Act 1992 to promote and facilitate investment within Papua New Guinea and abroad. The act regulates registration of companies/businesses and the certification of foreign enterprise. The Act states that investors are required to obtain licences for their area of business interest from the relevant Government agencies in the country.

The project is a mechanism for foreign investment in Papua New Guinea's sustainable forests and community development. The project developer has obtained the appropriate licences to operate in their area of businesses.

- **Industrial Relations Act (1962)**

This Act details the settlement of industrial disputes. It is an Act relating to development of industrial relations and the prevention and settlement of industrial disputes.

The project is aligned with this legislation through its own Grievance Redress processes including a formal dispute resolution strategy.

- **Industrial Health Safety and Welfare Act (1962)**

This Act relates to industrial safety, health and welfare, and related purposes. It ensures that measurements have been undertaken to ensure the health, safety, and welfare of employees in the industrial sector.

The project will take all necessary steps to ensure the working health and safety procedures for all the staff involved in the proposed activities.

- **Employment Act No. 54 of 1978 (Consolidated up to 31 March 2001)**

This Act relates to the employment of certain persons and lays out conditions of employment.

The project has employment policies and procedures that comply with the Act.

- **Child Welfare Act (1961)**

The welfare of children and related purposes are covered under this Act, it covers the protection and promotion of the rights and well-being of a child. Project objectives, including activities aimed at improving health standards and improving educational opportunities, are consistent with the Child Welfare Act.

The Project is committed to recognising the rights and welfare of children within the project area.

- **Discriminatory Practices Act (1963)**

This act is to prohibit certain discriminatory practices. 'Discriminatory practice' in this Act means discrimination either of an adverse or of a preferential kind, practiced by a person or group of persons, against or in favour of another person or group of persons for reasons only of colour, race or ethnic, tribal, or national origin.

The project promotes equal opportunities in the workplace based on personal merit rather than gender, colour, race or ethnic, tribal, or national origin.

- **Workers Compensation Act (1978)**

This Act provides for compensation to workers and their dependants in respect of injuries suffered by workers arising out of or in the course of their employment, and for related purposes.

The project is aligned with this act regarding employment policies and procedures related to workers compensation.

- **Papua New Guinea Vision (2050)**

This document maps out the future direction for Papua New Guinea's socio-economic development. It is underpinned by seven 'Strategic Focus Areas', which are referred to as pillars:

- Human Capital Development, Gender, Youth and People Empowerment;
- Wealth Creation;
- Institutional Development and Service Delivery;
- Security and International Relations;
- Environmental Sustainability and Climate Change;
- Spiritual, Cultural and Community Development; and
- Strategic Planning, Integration and Control

In alignment with Papua New Guinea 2050 Vision, the project will develop environmental and sustainability practices and social development and empowerment to the local communities.

## 2.4 Grouped Projects

This project is a grouped project activity.

New project activities are assessed and evaluated to determine if they provide significant individual impacts (funding of education), household (cookstove), or community level (training on climate change).

This monitoring period continues the 1st Project Instance (PI) that was included into the Grouped Project Activity (GPA) during the first monitoring period. There are no additional project instances that were included in this monitoring period.

# 3 BENEFITS FOR PEOPLE AND PROSPERITY

## 3.1 Impacts on Stakeholders

|                               |                          |
|-------------------------------|--------------------------|
| Impact #1                     | Provide access to ICS    |
| Type of Impact                | Positive, actual, direct |
| Affected Stakeholder Group(s) | ICS end-users            |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Resulting Change in Well-being</p> | <p>Describe characteristics and magnitude of impact on each stakeholder group</p> <p><b>Baseline:</b> The project activity aims to replace 100,000 traditional cookstoves with fuel efficient ICSs.</p> <p>Upon gaining access to the ICS, inadequate cooking will be gradually reduced. As will the burden of firewood collection, the time to prepare food and the cost to buy firewood, hence improving health and well-being.</p> <p><b>Monitoring Period:</b> During this monitoring period, an additional 497 traditional cookstoves were replaced with fuel efficient ICSs, bringing the total deployed ICSs to 19,703.</p>                                                                                                                                                                                                                          |
| <p>Impact #2</p>                      | <p>Improved Health Status</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p>Type of Impact</p>                 | <p>Positive, predicted, direct</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>Affected Stakeholder Group(s)</p>  | <p>ICS end-users</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p>Resulting Change in Well-being</p> | <p>Describe characteristics and magnitude of impact on each stakeholder group</p> <p><b>Baseline:</b> Prior to the project activity, high levels of smoke in the cooking place were observed during the pilot study which could lead to smoke borne illnesses.</p> <p>Through the use of the ICS, it gradually reduces the local people's exposure to high PM2.5 and high CO due to higher efficiency of combustion within their household over the lifetime of the project activity.</p> <p>The monitored baseline emission level from the open fire traditional cooking is 943.3 mg/MJ while the emission levels from newly distributed ICS is 177 mg/MJ.<sup>15</sup></p> <p><b>Monitoring Period:</b></p> <p>During this monitoring period, a total of 19,703 ICSs were operational, of which 497 were newly distributed in this monitoring period.</p> |

<sup>15</sup> [http://catalog.cleancookstoves.org/stove\\_details.html?stove\\_id=stove\\_AWOUJ](http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_AWOUJ)

|  |                                                                                                                                                                                                                                                                                                                                                                    |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Overall, there was a total decrease of 1,328.06 tonnes of PM2.5 from the usage of the ICSs.</p> <p>Additionally, this is further supported by the Ground Monitoring Survey in which 100% of households reported observing only thin and light smoke and 94% of households reporting a decrease in intensity of the smoke compared to the baseline scenario.</p> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Impact #3                      | Less time spent on unpaid domestic and care work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Type of Impact                 | Positive, predicted, direct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Affected Stakeholder Group(s)  | ICS end-users                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Resulting Change in Well-being | <p>Describe characteristics and magnitude of impact on each stakeholder group</p> <p><b>Baseline:</b> Due to higher thermal efficiency, less firewood is required during the cooking process, and time spent cooking is significantly reduced. This results in a direct reduction in time spent on firewood collection and cooking for people.</p> <p>In the baseline, it was found that end-users would collect firewood 3-5 times/week, spending on average 1.5 hours (6.25%) a day. Upon sampling during the monitoring period, this frequency of collection should reduce as less firewood is required for cooking and heating throughout the week.</p> <p><b>Monitoring Period:</b> During this monitoring period, during sampling, monitoring of the frequency of collection of firewood was done. The frequency decreased to 1 time/week, an average of 0.43 hours (1.78) per day which is lower than the baseline.</p> <p>This decrease from the baseline scenario results in total time saved on un-paid work to be 1.07 hours per day.</p> |

|                |                          |
|----------------|--------------------------|
| Impact #4      | Increase Income          |
| Type of Impact | Positive, actual, direct |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Affected Stakeholder Group(s)  | Local residents, Employees hired for Project Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Resulting Change in Well-being | <p>Describe characteristics and magnitude of impact on each stakeholder group</p> <p><b>Baseline:</b> During the project crediting period, TEM will oversee maintenance of ICSs and the monitoring plan, which will require the need to hire local people.</p> <p>Local people will be hired through a fair and unbiased process, during which we estimate that 130 jobs (10 full-time and 120 part-time) will be created over the project lifespan. The Project targets hiring females as at least 50% of all employees. Employees will be compensated above the average income of 439 Kina /month<sup>16</sup>.</p> <p><b>Monitoring Period:</b> During this current monitoring period, 6 additional people were onboarded for the project, for a total of 30 local staff hired for the project (2 full-time and 28 part-time).</p> <p>These part-time employees were compensated with an average income of (1,369.84 Kina / month) and is pro-rated where part-time employment is relevant.</p> |
| Impact #5                      | Trainings imparted on climate change, project implementation and monitoring procedures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Type of Impact                 | Positive, actual, direct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Affected Stakeholder Group(s)  | Local community, Beneficiaries, TEM in-country team and Staff                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

<sup>16</sup> Bonilla, F., Voorend, K., Anker, R., Anker, M., & Prates, I. (2020). Anker Living Income Reference Value: Rural Papua New Guinea (2020) (No.20-03-20). Universidad Privada Boliviana.

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Resulting Change in Well-being</b></p> | <p>Describe characteristics and magnitude of impact on each stakeholder group</p> <p><b><u>Baseline:</u></b> Prior to the project activity, local residents and end-users had little awareness of ICSs and climate change.</p> <p>With the inception of the project activity, training and skill development related to community engagement, survey conduction, technical trainings like conducting Water Boiling Tests (WBT) will be provided to many stakeholder groups which is envisaged to empower their lives by not only providing long-term employability but also through increased awareness levels regarding issues related to climate change, social equity.</p> <p>This will be done during the onboarding of new staff throughout the lifespan of the project activity.</p> <p><b><u>Monitoring Period:</u></b> During this monitoring period, local people who were hired were put through training and skill development as mentioned above.</p> <p>Training attendances and materials are present which can be made available to the validator upon request. A total of <b>30</b> local staff (<b>2</b> full-time positions and <b>28</b> part-time positions) were trained for their respective roles. All staff are trained annually to ensure that knowledge transferred to them is up to date and the delivery of the project remains on high standards.</p> |
| <p><b>Impact #6</b></p>                      | <p>Improved Economic Outcomes</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Type of Impact</b></p>                 | <p>Positive, actual, direct</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Affected Stakeholder Group(s)</b></p>  | <p>ICS end-users, ICS end-users' Families</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Resulting Change in Well-being</p> | <p>Describe characteristics and magnitude of impact on each stakeholder group</p> <p><b>Baseline:</b> With the commencement of the project activity, the use of the ICS increases fuel efficiency, and an increase in monetary savings is expected. This change is the result of the reduction in the cost of firewood purchases over the lifespan of the project activity due to the increase in fuel efficiency with the ICSs.</p> <p>As per the project activity's Emission Reduction Calculations for project activities in the Highlands, the quantity of firewood calculated to be saved was 3.7 tonnes/year. Coupled with the Baseline Survey Report where it was found that end-users on average spent K540/year on firewood.</p> <p><b>Monitoring Period:</b> During this monitoring period, the project has resulted in savings of 3.7 tonnes of non-renewable biomass per year per ICS as well as a reduction in trips taken per week to collect or buy firewood.</p> <p>Additionally, it was recorded in the monitoring survey that approximately 6% of the end-users buy firewood. The project has resulted in a total cost saved of 1.04 Kina per household per day. The savings per household per day is estimated to be 1.04 Kina for households that purchase their own firewood. This is consistent with the responses from the monitoring survey, where 100% of households buying firewood reported an average savings of 5-10 Kina per week. By proportioning the number of households that purchase their firewood from the monitoring survey, 107,027 Kina is saved during this period across all households.</p> <p>Overall, the project ultimately results in long-term household savings during this monitoring period.</p> |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                               |                                                     |
|-------------------------------|-----------------------------------------------------|
| Impact #7                     | Reduction in sales of Firewood for Firewood Vendors |
| Type of Impact                | Negative, actual, direct                            |
| Affected Stakeholder Group(s) | Firewood Vendors                                    |



### Resulting Change in Well-being

Describe characteristics and magnitude of impact on each stakeholder group

**Baseline:** In the absence of the project activity, end-users may opt to procure firewood in order to meet the volumetric need for daily cooking and heating practices.

With the commencement of the project activity, there is a possibility that there will be a reduction in demand of firewood associated with the ICSs that might affect the vendors selling fuel wood in the project area. However, considering the significant demand-supply gap on fuelwood for cooking needs, decrease in the fuel-wood demand in project location is not likely to have a significant negative impact on wood vendors.

**Monitoring Period:** During this monitoring period, it was determined that there could be a decrease in fuel wood consumption from the baseline (file titled: VCS4416\_TEM 2025\_Cookstoves Highlands region report\_v3.pdf) as mentioned in the decrease in cost spent by ICS end users in impact #6.

The majority of the rural population still rely on wood fuel, creating a demand-supply gap in the country. Hence, any reduction for wood-fuel within the project boundaries will shift stakeholders to sell wood outside the project boundary, without any significant reduction in their business.

## 3.2 Stakeholder Impact Monitoring

During this monitoring period, TEM's trained in-country team, conducted sampling to collect data through interviews, surveys, direct observations, and group discussions regarding stakeholders' health and employment records.

The sampling plan for the SD VISta project activity has been designed based on the requirements of the allied VCS project activity – "Improved Cookstove Grouped Project in Papua New Guinea".

TEM's field teams gather all necessary information to identify households using its ICS during the project. To facilitate this process, each ICS is assigned a Unique Serial Number (USN). This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- ICS model
- Date of distribution/installation

- ICS serial number
- Retailer/distributor information

Additionally, sampling is carried out for the monitoring of the project activity. The project activity distributes ICSs to households, which will have a net positive impact on the overall well-being of the people in the project area. To account for these positive impacts, the sampling plan identifies the parameters and sampling methods required.

The parameters monitored for the relevant activities with benefit for People and Prosperity and the respective evidence is mentioned in *Table 7* below.

**Table 7:** Parameters monitored for activities related to Benefits for People and Prosperity

| S/N | Parameter                                           | Unit   | Description                                                                                                              | Evidence Mode                                                                       |
|-----|-----------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1   | Number of ICSs distributed                          | Number | Total number of ICSs distributed contributes to a reduction of firewood used, which is usually purchased at high prices. | Distribution Ledger                                                                 |
| 2   | Level of fine particulate matter (PM2.5) emissions. | mg/MJ  | Reduction in the level of fine particulate matter (PM2.5) emissions within households                                    | Clean Stove Catalogue <sup>17</sup>                                                 |
|     |                                                     |        | Reduction in the amount of smoke produced                                                                                | Ground Monitoring Survey                                                            |
|     |                                                     |        | Reduction in thermal efficiency in older cookstoves                                                                      | BURN study / Water Boiling Test as per VM0050 v1.0                                  |
| 3   | Number of participants during training sessions     | Number | Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex             | Signed participation/ attendance forms, onboarding documents, employment contracts. |
| 4   |                                                     |        | Number of youth and adults with technical and vocational skills, by type of skill within the project area.               |                                                                                     |

<sup>17</sup> [http://catalog.cleancookstoves.org/stove\\_details.html?stove\\_id=stove\\_AW0UJ](http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_AW0UJ)

|   |                                                       |        |                                                                                              |                                                |
|---|-------------------------------------------------------|--------|----------------------------------------------------------------------------------------------|------------------------------------------------|
| 5 | Employment contracts                                  | Number | Number of people employed, by sector and sex, under the project activity                     | Project Employee Breakdown and Analysis Report |
| 6 | Employee Pay slips                                    | Number | Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities | Project Employee Breakdown and Analysis Report |
| 7 | Number of trips made to collect/buy firewood per week | Number | Proportion of time spent on unpaid domestic and care work, by sex, age, and location.        | Ground Monitoring Survey                       |
| 8 | Number of women hired for the project activity        | Number | Number of women in the project including managerial roles.                                   | Female Employment contracts                    |

### 3.3 Net Positive Stakeholder Well-being Impacts

The inception and distribution of ICSs undertaken by the project activity will provide a basic service necessary to lead to a healthy and productive life, including saving time and money for wood fuel at the household level. The project will reduce the drudgery undertaken by them especially for women and children through time savings in cutting, collecting, and carrying firewood from trees.

Through the development of the project activity, local stakeholders will yield the following benefits:

1. **19,703** households were provided with access to the fuel-efficient ICS for meeting their daily cooking needs during the project lifetime.
2. Emissions of PM2.5 is reduced to be below the baseline emission level by **81.24%** to 177 mg/MJ in project households for ICSs. Older cookstoves provide a lower reduction. Overall, this results in a decrease in **1,328.06** tonnes of PM2.5 during this monitoring period. Additionally, this is further supported by the Ground Monitoring Survey in which 100% of households reported observing only thin and light smoke and 94% of households reporting a decrease in intensity of the smoke compared to the baseline scenario.
3. The proportion of time taken to collect firewood decreased from **6.25%** of a day in the baseline scenario to **1.78%** of a day during this monitoring period. This decrease resulted in a total time of **1.07 hours** saved per day per household.

4. An increase in the economic outlook and an improvement in the technical and vocational skills of local people as **30** local people were hired to be part of the project team. Out of the people hired, 28 were hired on a part-time basis with an average income of **1,369.84** Kina per month.
5. There would be an increase of local people who are trained on climate change, project implementation and monitoring procedures. During this monitoring period, a total of **30** individuals were trained with **8** being female and **22** being male.
6. Families will have a greater capacity to save money used for buying wood fuel due to the reduced demand of fuel wood for cooking purposes. For end-users who buy firewood, this has resulted in total cost savings of 1.04 Kina per household per day. By proportioning the number of households that purchase their firewood from the monitoring survey, **107,027 Kina** is saved during this period in households that buy firewood. This is consistent with the responses from the monitoring survey, where 100% of households buying firewood reported an average savings of 5-10 Kina per week.

## 4 BENEFITS FOR THE PLANET

### 4.1 Impacts on Natural Capital and Ecosystem Services

|                                                      |                                                                                                                                                        |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Impact #1                                            | Reduced demand for large diameter non-renewable firewood from live trees due to only needing finger-sized woody biomass, i.e., twigs and crop residues |
| Type of Impact                                       | Positive, actual, direct                                                                                                                               |
| Affected Natural Capital and/or Ecosystem Service(s) | Live plant biomass, Biodiversity and species Richness, Soil and Water Conservation                                                                     |

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Resulting Change in Condition</p>                        | <p>Describe characteristics and magnitude of impact on each affected natural capital and ecosystem service</p> <p><b>Baseline:</b> By slowing the consumption of woody biomass for cooking purposes and by eliminating the need for large diameter woody biomass due to the technology in the ICSs, the resulting drop in demand slows deforestation, thus increasing the above-ground biomass in the adjoining forests near the project area. Reduction in unsustainable harvesting of plant biomass would in turn reduce soil erosion and result in improvement in biodiversity and species richness of the implementation area.</p> <p><b>Monitoring Period:</b></p> <p>During this monitoring period, both the frequency of firewood collection and the amount spent on firewood has decreased. This drop in demand would slow down deforestation and thus avoiding the loss of the above-ground biomass in the adjoining forests near the project area</p> <p>The project has resulted in savings of <b>3.7</b> tonnes of non-renewable biomass per year per ICS. Over this monitoring period, a total savings of <b>73,470</b> tonnes of biomass from the <b>19,703</b> ICS in operation.</p> |
| <p>Impact #2</p>                                            | <p>Avoided emission of GHGs made possible by using ICSs</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>Type of Impact</p>                                       | <p>Positive, predicted, indirect</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>Affected Natural Capital and/or Ecosystem Service(s)</p> | <p>Air quality in the households as well as surroundings</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Resulting Change in Condition

Describe characteristics and magnitude of impact on each affected natural capital and ecosystem service

**Baseline:** The ICSs require less fuelwood for producing the same amount of thermal energy, therefore, resulting in less carbon dioxide emission in the atmosphere. As carbon dioxide is a greenhouse gas, its reduced emission translates to a positive impact on climate change.

#### **Monitoring Period:**

During this monitoring period, a total of 73,189 tCO<sub>2</sub> was estimated to be reduced due to the inception of the project activity.

## 4.2 Natural Capital and Ecosystem Services Impact Monitoring

Similarly to section **Error! Reference source not found.**, the impact on natural capital and ecosystem services were monitored by TEM's in-country team. The data was collected through surveys and kitchen performance tests (KPTs). The data from tests and surveys were recorded, analysed and reported by TEM's field team. The parameters monitored for the impacts relevant to benefits for the planet with its respective evidence during this monitoring period are summarised in *Table 8* below.

**Table 8: Monitoring plan for activities with benefit for the Planet**

| S/N | Parameter                                                      | Unit             | Description                                                                                                             | Evidence Mode                                                                                          |
|-----|----------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 1   | Number of ICSs distributed during the monitoring period        | Number           | Total number of ICS distributed contributes to a reduction of firewood used, which is usually purchased at high prices. | Distribution Ledger                                                                                    |
| 2   | Tonnes of CO <sub>2</sub> removed during the monitoring period | TCO <sub>2</sub> | Tonnes of greenhouse gas emissions avoided or removed                                                                   | Kitchen Performance Tests (KPTs)<br>&<br>Baseline Surveys<br>&<br>Emission Reduction Calculation Sheet |
| 3   | Tonnes of biomass saved during the monitoring period           | Tonnes           | Progress towards sustainable forest management through the increase of above-ground biomass in forests                  |                                                                                                        |

### 4.3 Net Positive Natural Capital and Ecosystem Services Impacts

By replacing traditional low efficient traditional open fire cooking with high efficiency ICSs in households, the project will increase energy efficiency resulting in less firewood combustion, thus generate net GHG reductions.

The average annual GHG emissions reduction from the project is expected to be **320,099 tCO<sub>2</sub>e**. The crediting period is expected to be 10 years. Hence the total GHG emissions reduction is **3,200,995 tCO<sub>2</sub>e**. The project will save 3.7 tonnes of non-renewable biomass every year per ICS and it will result in a significant reduction of deforestation.

For this monitoring period, the GHG emissions reduced were **73,189 tCO<sub>2</sub>e** and a total of **73,470 tonnes of non-renewable biomass** were saved.

# APPENDIX I: STAKEHOLDER CONSULTATIONS

During the current monitoring period, the following stakeholder interactions in SHP were conducted:

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ongoing consultation</b>                | Awareness meetings and demonstrations on cookstove benefits.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Date(s) of stakeholder consultation</b> | 25 March 2025 (East Pangia)<br>30 June 2025 (Imbonggu, Mendi, Ialibu)<br>11 September 2025 (Western Pacific University)<br>12 October 2025 (Ialibu)<br>17-18 December 2025 (Mendi, Ialibu, Kewabi, Imbonggu)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Communication of monitored results</b>  | Throughout the implementation of the project, awareness programs were conducted on the benefits of ICSs in multiple wards and villages.<br><br>A series of targeted meetings and live demonstrations were organised to highlight the benefits of the cookstoves. These sessions were designed to be interactive, allowing attendees to see first-hand how the cookstoves operated and to ask questions directly to our facilitators. Such demonstrations effectively addressed common concerns and misconceptions, making the technology more accessible and relatable, while boosting behavioural change. On going awareness activities strengthened long-term engagement among improved cookstove users, enabling them to realise the full advantages of the technology throughout numerous wards and villages. |
| <b>Consultation records</b>                | Photographs and staff records                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Stakeholder input</b>                   | The activities were actively attended by end-users who acknowledged the benefits of the improved cookstoves along with best practices on using and maintaining them.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |



|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Ongoing consultation                | District level awareness and consultation meetings with community leaders                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Date(s) of stakeholder consultation | 22 April 2025 (Pangia District)<br>23 April 2025 (Kagua District)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Communication of monitored results  | <p>Prior to the distribution of ICSs, consultations meetings are carried out to inform on the benefits of ICSs in every district.</p> <p>Each session is designed to inform and engage community leaders on the Improved Cookstoves Project. During these sessions the project's benefits were explained, community concerns were addressed, while fostering collaboration for successful implementation in the Southern Highlands Province. TEM's technical team leading the awareness sessions introduced the field team members and their roles to the locals in Pangia and Kagua Districts and explained the steps needed to be involved in the project. The sessions ended with an open forum where the participants were encouraged to ask questions and share feedback.</p> |
| Consultation records                | Attendance sheets of participants, copies of support letters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Stakeholder input                   | Participants recognised the initiative as a much-needed service, particularly for mothers in remote communities who often face the greatest challenges. The project was acknowledged not only for its cost-saving potential but also for its significant health and environmental benefits, reinforcing strong community support.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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| Ongoing consultation                | Awareness and Household Registration of households to take part in the ICS program.                                                                                                                                                                                                              |
| Date(s) of stakeholder consultation | 25 March 2025 – 27 March 2025                                                                                                                                                                                                                                                                    |
| Communication of monitored results  | <p>Prior to the distribution of ICSs, awareness sessions were conducted on the benefits of ICSs in every ward and village.</p> <p>Households interested in taking part in the project were registered through TEM's field assistants using their personal details and household information.</p> |

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|                      | Households who met the household selection criteria were onboarded into the project and were given a token flyer to exchange for an ICS.                                                                                                      |
| Consultation records | Information was recorded in documents by software to keep track of the ownership of the ICS.<br><br>A total of 497 household registrations were done.                                                                                         |
| Stakeholder input    | Interested households provided information such as its geographical location and contact details of the head of the family or end-user. Consent form/Household Registration form were also signed off by the head of the family or end-users. |

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| Ongoing consultation                | Collection of an ICS by end-users who have undergone Household Registration previously                                                                                                                                                                                                                                                                                                                                 |
| Date(s) of stakeholder consultation | 28 March 2025                                                                                                                                                                                                                                                                                                                                                                                                          |
| Communication of monitored results  | End-users previously registered brought their token flyers to the distribution sites where they were matched to our database and collected by the distribution team.<br><br>Collected tokens were exchanged for an ICS.                                                                                                                                                                                                |
| Consultation records                | A data entry officer was present to manage data collected by the distribution team.<br><br>A total of 497 tokens flyers were collected and exchanged for an ICS.                                                                                                                                                                                                                                                       |
| Stakeholder input                   | All information provided was collected and managed by the data entry officer and the warehouse and logistic officer who manages the data collected from the households. Information regarding end-users would be updated once they have received the ICS.<br><br>The information collected will include: <ul style="list-style-type: none"> <li>• Name of ICS user,</li> <li>• Address/GPS of ICS household</li> </ul> |

|  |                                                                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• Phone number of ICS user</li> <li>• ICS model</li> <li>• Date of distribution and installation</li> <li>• Name of staff that delivered and installed the ICS</li> <li>• Photographic Evidence</li> </ul> |
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| Ongoing consultation                | Presentation of key project benefits of the project to PNG Resource Week, a national forum in Port Moresby                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Date(s) of stakeholder consultation | 20 July 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Communication of monitored results  | <p>During PNG Resource Week in Port Moresby, the Day 2 program on Impact in Renewables and Climate Change, held on 11 September 2024, featured a detailed presentation by Dr. Gabriel Machovsky-Capuska on the VCS 4416 PNG Cookstoves Project. The audience included government energy officials, provincial administrators, private-sector actors, development agencies, and community organizations working to advance clean-energy solutions.</p>  <p>The presentation outlined how the project seeks to reduce indoor pollution by promoting efficient cookstoves designed for PNG's diverse cultural and geographic settings. He shared scientific findings on existing cooking behaviours, highlighting associated health concerns, fuel-use patterns, and the environmental burden of biomass burning. He further emphasized the importance of participatory design,</p> |

|                      |                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | reliable distribution systems, and continuous monitoring to ensure the stoves remain practical and widely used.                                                                            |
| Consultation records | Photographs of the event                                                                                                                                                                   |
| Stakeholder input    | The session ended with a strong appeal for a coordinated national collaboration to scale clean-cooking technologies and strengthen sustainable resource management throughout the country. |

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|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ongoing consultation                | <p>A day of training was conducted for TEM's in-country field assistants before conducting the monitoring survey.</p> <p>This was followed by three days of ground monitoring surveys, conducted for this monitoring period's end-users as per the sampling plan.</p>                                                                                 |
| Date(s) of stakeholder consultation | 05-Jan-2026                                                                                                                                                                                                                                                                                                                                           |
| Communication of monitored results  | Training workshops were conducted in Tok Pidgin, the native language of Papua New Guinea, by TEM's in-country leadership team. During this workshop, field assistants reviewed the importance of collecting data accurately, while respecting the privacy of the end-users and were briefed on the logistics and overall importance of these surveys. |
| Consultation records                | <p>Results from the ground monitoring survey were collected and collated to be used in the ER calculation.</p> <p>A total of 150 households were selected, and 149 Sampling Surveys were successfully completed between the dates 06-08 January 2026.</p>                                                                                             |
| Stakeholder input                   | All inputs which were required by the sampling plan was collected from end-users through measurements, observations and a list of survey questions.                                                                                                                                                                                                   |